

CONFIGURING BGP WITH OSPF, IS-IS AND EIGRP 2025



BGP

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Purpose:

The purpose of this lab was to configure three ASes, running OSPF, EIGRP, and IS-IS respectively, and then connect them using BGP.

Background:

Border gateway protocol is the dominant outer gateway protocol in use throughout the internet. However, BGP was not the first protocol of its kind that would try and bridge the internet.

One of the first gateway protocols is the Gateway-to-gateway protocol. Developed in 1982, it used minimized the amount of ASes in routes. After that, the backbone of the internet became EGP, developed in the early 1980s by Bolt, Beranek, and Newman. However, it had some major limitations. First of all, as a simple reachability protocol, it is limited to tree-like topologies and does not support multi path networking. In addition, because it was designed to be centrally controlled, which is a major problem for the modern internet, because there is no central authority to manage it. To address the problems with EGP, a new routing protocol was created. BGP was developed in 1989 by Kirk Lougheed, Len Bosack, and Yakov Rekhter. New versions were quickly released afterwards, with BGP-2 coming out in June 1990, BGP-3 coming out in October 1991 and BGP-4 coming out July 1994.

Routing protocols are generally separated into three classes, distance-vector protocols, link-state protocols, and path-vector protocols. Distance vector protocols operate by exchanging information about the distance between to various destinations between various nodes, and link state protocols requires each node to maintain a complete map of the network topology. Path vector protocols transmit both distance and the entire path to a destination. None of the distance vector and link state protocols developed up to the point of BGP had the flexibility to deal with the intricacies of the internet.

BGP works by splitting the internet into different autonomous systems (AS), and then routing between them. These autonomous systems use internal gateway protocols like OSPF and EIGRP, a link-state and distance vector protocol respectively, to provide internal routing in the AS. Thus, the role of BGP is to connect these ASes.

In order for BGP to function, the border routers have an active BGP session between them. Once connected, BGP update packets scappy a full list of transit autonomous systems that must be traversed between the transmitting and receiving AS. In order to prevent loops, BGP drops any paths where the AS path list contains itself. This method of transmitting routes is what gives bgp the flexibility that it needs to route the internet. Because of the additional information that can be passed through these transit lists, BGP can store backup routes, effectively avoid untrustworthy sites, and many other features that give it the flexibility it needs to route the complicated global internet.

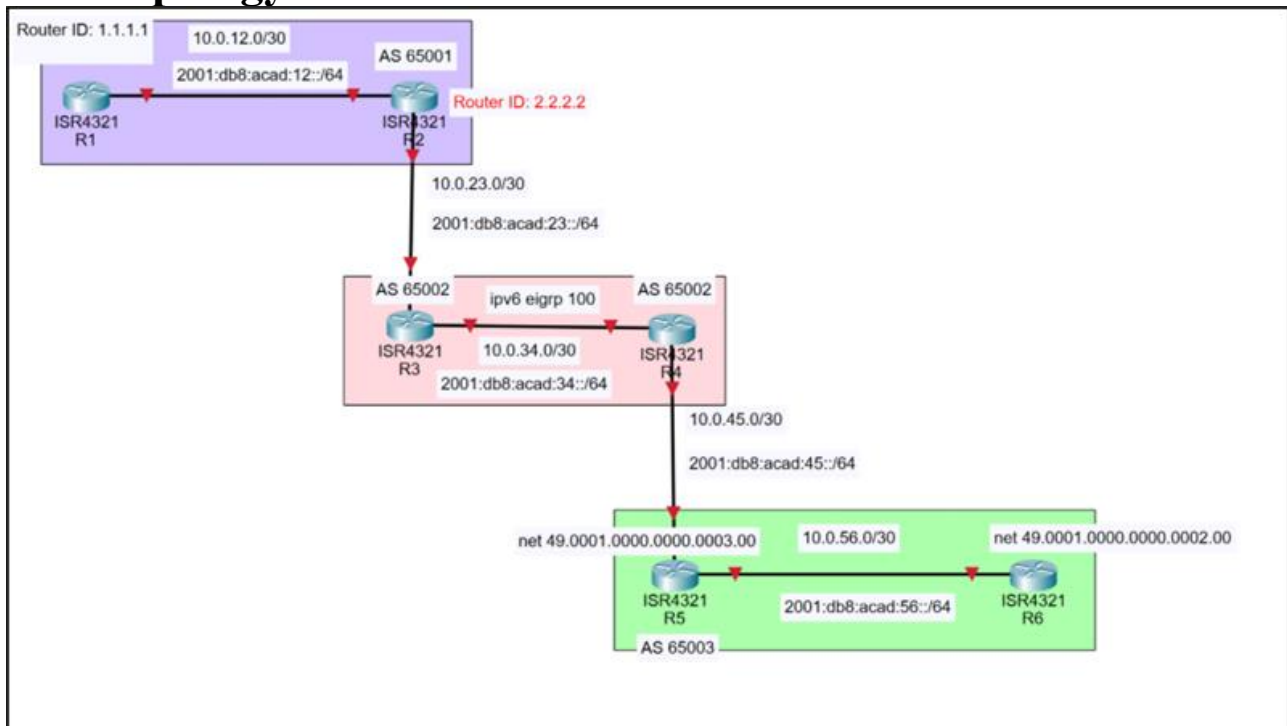
These routers have critical importance to the internet, as misconfiguring them can lead to entire sections of the world to lose connectivity. For example, in 2008, a Pakistani telecom company announced new more specific routes to youtube.com, that would than black hole the traffic. Doing so would block YouTube for the citizens of Pakistan However, the ISP than advertised these new routes upstream, blocking YouTube for a large portion of the global internet. These misconfigurations are often cited as a major cause when the internet goes down.

Lab Summary:

In this lab, six routers were set up in a line topology, and every two routers were assigned an AS: AS 65001, 65002, and 65003 respectively. Each of these ASes ran a different internal gateway

protocol. AS 65001 ran OSPF, AS 65002 ran EIGRP, and AS 65003 ran IS-IS. We then set up external BGP on the edge routers, and configured the Weight and AS_PATH attributes on router three.

Lab Topology:



Lab Commands:

eBGP Commands:

`router bgp <1-65535>` Autonomous system number

Enables BGP processes on the route

`bgp router-id A.B.C.D` Manually configured router identifier

Assigns a router-id to the bgp process

`bgp log-neighbor-changes`

Logs neighbor changes

`neighbor A.B.C.D remote-as <1-65535>`

establishes a bgp peer session with a neighbor on the specified as with the specified ip address

`neighbor 2001:DB8:ACAD:23::2 remote-as 65002`

establishes a bgp peer session with a neighbor on the specified as with the specified ip address

`address-family ipv4`

enters into ipv4 config mode for bgp

`network 10.0.23.0 mask 255.255.255.252`

Distributes neighboring network into bgp

`redistribute eigrp 100 metric 10`

Distributes eigrp routes from eigrp process specified into BGP with specified

`redistribute connected`

Redistributes all directly connected routes into bgp

`redistribute ospfv3 1 metric 10`

Redistributes ospfv3 routes into BGP with metric 10

`neighbor 10.0.23.2 activate`

Activates neighboring router to start bgp session

`neighbor 10.0.23.2 route-map SET_WEIGHT in`

Assigns a route map to a bgp connection

`redistribute isis CORE level-2 metric 10`

Redistributes all level-2 routes from the isis CORE group into bgp

`address-family ipv6`

enters into ipv6 address family configuration mode

`redistribute connected`

Redistributes directly connect ipv6 networks

`redistribute ospf 1 metric 10 match internal external 1
external 2`

Redistributes both internal and external ospf routes into bgp with a metric of 10

`network 2001:DB8:ACAD:23::/64`

Distributes Specified network into bgp

`neighbor 2001:DB8:ACAD:23::2 activate`

Activates BGP session with bgp neighbor

`neighbor 2001:DB8:ACAD:23::2 next-hop-self`

Forces bgp to add itself as the next hop address when talking to specified iBGP neighbor

`ip prefix-list ALL_ROUTES seq 5 permit 0.0.0.0/0 le 32`

Creates a prefix list that matches all ip addresses

`route-map SET_WEIGHT permit 10`

Creates a route map that permits traffic

`set weight 249`

Sets the weight attribute for this route map

`match ip address prefix-list ALL_ROUTES`

Matches routes to the ip prefix-list specified

`set local-preference 150`

Sets the local pref value for the route map

OSPFv3 commands:

`ospfv3 1 ipv4 area 0`

Assigns an interface to ospfv3 process an a ipv4 area

`ospfv3 1 ipv6 area 0`

Assigns an interface to ospfv3 process an a ipv6 area

`router ospfv3 1`

Starts ospfv3 processes with the specified process id

`router-id 2.2.2.2`

Assigns a router id to this process

`address-family ipv4 unicast`

Enters into ipv4 unicast config mode for ospfv3

`redistribute connected`

Redistributes directly connected routes into ipv4 ospf with metric 10

`redistribute bgp 65001 metric 10`

Redistributes bgp routes into ipv4 ospf with metric 10

`address-family ipv6 unicast`

Enters into ipv6 unicast config mode for ospfv3

`redistribute connected`

Redistributes directly connected routes into ipv6 ospf with metric 10

`redistribute bgp 65001 metric 10`

Redistributes bgp routes into ipv6 ospf with metric 10

EIGRP Commands:

`ipv6 eigrp 100`

Adds an interface into ipv6 eigrp

`router eigrp 100`

Start ipv4 eigrp processes on the router, with the PID 100

`eigrp router-id 3.3.3.3`

Assigns router id to ipv6 router process

```
network 3.3.3.3 0.0.0.0
```

Adds the specified network into eigrp routing

```
redistribute bgp 65002 metric 100 1 255 1 1500
```

Redistributes bgp routes into ipv4 eigrp with the metrics Bandwidth, Load, Delay, and Reliability set, respectively

```
redistribute connected
```

Redistributes directly connected routes into ipv4 eigrp

```
ipv6 router eigrp 100
```

Start ipv6 eigrp processes on the router

```
eigrp router-id 3.3.3.3
```

Assigns router id to ipv6 router process

```
redistribute bgp 65002 metric 10000 100 255 1 1500
```

Redistributes bgp routes into ipv6 eigrp with the metrics Bandwidth, Load, Delay, and Reliability set, respectively

```
redistribute connected
```

Redistributes directly connected routes into ipv6 eigrp

IS-IS commands:

```
router isis CORE
```

Starts isis processes on router on isis process CORE

```
is-type level-2-only
```

Turns is-is processes to only route level 2 is-is routes

```
net 49.0001.0000.0000.0003.00
```

Assigns net id to is-is process

```
redistribute connected metric 10
```

Redistributes directly connected routed into is-is ipv4

```
redistribute bgp 65003 metric 10
```

Redistributes bgp routes into is-is ipv4

```
address-family ipv6
```

Enters into ipv6 config mode

```
redistribute connected metric 10
```

Redistributes directly connected routed into is-is ipv6

```
redistribute bgp 65003 metric 10
```

Redistributes bgp routes into is-is ipv6

```
ip router isis CORE
```

Enables ip isis routing on an interface in isis process CORE

```
ipv6 router isis CORE
```

Enables ipv6 is-is routing on an interface in isis process CORE

Lab Commands:

R1:

```
interface Loopback0
ip address 1.1.1.1 255.255.255.255
ipv6 address 2001:DB8:ACAD::1/128
ipv6 enable
ospfv3 1 ipv6 area 0
ospfv3 1 ipv4 area 0
interface GigabitEthernet0/0/0
ip address 10.0.12.1 255.255.255.252
negotiation auto
ipv6 address 2001:DB8:ACAD:12::1/64
ipv6 enable
ospfv3 1 ipv4 area 0
ospfv3 1 ipv6 area 0
interface GigabitEthernet0/0/1
no ip address
negotiation auto
interface GigabitEthernet0
vrf forwarding Mgmt-intf
no ip address
shutdown
negotiation auto
router ospfv3 1
router-id 1.1.1.1
address-family ipv4 unicast
exit-address-family
address-family ipv6 unicast
```

```
exit-address-family
ip forward-protocol nd
ip http server
ip http authentication local
ip http secure-server
ip tftp source-interface GigabitEthernet0
control-plane
line con 0
transport input none
stopbits 1
line aux 0
stopbits 1
line vty 0 4
login
```

R2:

```
interface Loopback0
ip address 2.2.2.2 255.255.255.255
ipv6 address 2001:DB8:ACAD::2/128
ipv6 enable
ospfv3 1 ipv4 area 0
ospfv3 1 ipv6 area 0
interface GigabitEthernet0/0/0
ip address 10.0.12.2 255.255.255.252
negotiation auto
ipv6 address 2001:DB8:ACAD:12::2/64
ipv6 enable
ospfv3 1 ipv4 area 0
ospfv3 1 ipv6 area 0
interface GigabitEthernet0/0/1
ip address 10.0.23.1 255.255.255.252
negotiation auto
ipv6 address 2001:DB8:ACAD:23::1/64
ipv6 enable
interface Serial0/1/0
```



```
no ip address
shutdown
interface Serial0/1/1
no ip address
shutdown
interface GigabitEthernet0
vrf forwarding Mgmt-intf
no ip address
shutdown
negotiation auto
router odr
router ospfv3 1
router-id 2.2.2.2
address-family ipv4 unicast
    redistribute connected
    redistribute bgp 65001 metric 10
exit-address-family
address-family ipv6 unicast
    redistribute connected
    redistribute bgp 65001 metric 10
exit-address-family
router bgp 65001
bgp router-id 2.2.2.2
bgp log-neighbor-changes
no bgp default ipv4-unicast
neighbor 10.0.23.2 remote-as 65002
neighbor 2001:DB8:ACAD:23::2 remote-as 65002
address-family ipv4
    network 10.0.23.0 mask 255.255.255.252
    redistribute connected
    redistribute ospfv3 1 metric 10
    neighbor 10.0.23.2 activate
    neighbor 10.0.23.2 route-map SET_WEIGHT in
exit-address-family
```

```
address-family ipv6
  redistribute connected
  redistribute ospf 1 metric 10 match internal external 1 external 2
  network 2001:DB8:ACAD:23::/64
  neighbor 2001:DB8:ACAD:23::2 activate
  neighbor 2001:DB8:ACAD:23::2 next-hop-self
exit-address-family
ip forward-protocol nd
ip http server
ip http authentication local
ip http secure-server
ip tftp source-interface GigabitEthernet0
route-map SET_WEIGHT permit 10
set weight 250
control-plane
line con 0
transport input none
stopbits 1
line aux 0
stopbits 1
line vty 0 4
login
end searcheng
```

R3:

```
interface Loopback0
ip address 3.3.3.3 255.255.255.255
ipv6 address 2001:DB8:ACAD::3/128
ipv6 enable
ipv6 eigrp 100
interface GigabitEthernet0/0/0
ip address 10.0.34.1 255.255.255.252
negotiation auto
ipv6 address 2001:DB8:ACAD:34::1/64
```

```
ipv6 enable
ipv6 eigrp 100
interface GigabitEthernet0/0/1
ip address 10.0.23.2 255.255.255.252
negotiation auto
ipv6 address 2001:DB8:ACAD:23::2/64
ipv6 enable
interface Serial0/1/0
no ip address
shutdown
interface Serial0/1/1
no ip address
shutdown
interface GigabitEthernet0
vrf forwarding Mgmt-intf
no ip address
shutdown
negotiation auto
router eigrp 100
network 3.3.3.3 0.0.0.0
network 10.0.34.0 0.0.0.3
redistribute bgp 65002 metric 100 1 255 1 1500
redistribute connected
router eigrp 65002
router bgp 65002
bgp router-id 3.3.3.3
bgp log-neighbor-changes
neighbor 10.0.23.1 remote-as 65001
neighbor 2001:DB8:ACAD:23::1 remote-as 65001
address-family ipv4
    network 10.0.23.0 mask 255.255.255.252
    redistribute connected
    redistribute eigrp 100
    neighbor 10.0.23.1 activate
```

```
neighbor 10.0.23.1 route-map SET_LOCALPREF in
no neighbor 2001:DB8:ACAD:23::1 activate
exit-address-family
address-family ipv6
    redistribute connected
    redistribute eigrp 100 metric 10
    network 2001:DB8:ACAD:23::/64
    neighbor 2001:DB8:ACAD:23::1 activate
    neighbor 2001:DB8:ACAD:23::1 next-hop-self
exit-address-family
ip forward-protocol nd
ip http server
ip http authentication local
ip http secure-server
ip tftp source-interface GigabitEthernet0
ip prefix-list ALL_ROUTES seq 5 permit 0.0.0.0/0 le 32
ipv6 router eigrp 100
eigrp router-id 3.3.3.3
redistribute bgp 65002 metric 10000 100 255 1 1500
redistribute connected
route-map SET_WEIGHT permit 10
set weight 249
route-map SET_LOCALPREF permit 10
match ip address prefix-list ALL_ROUTES
set local-preference 150
control-plane
line con 0
transport input none
stopbits 1
line aux 0
stopbits 1
line vty 0 4
login
end
```

R4:

```
interface Loopback0
ip address 4.4.4.4 255.255.255.255
ipv6 address 2001:DB8:ACAD::4/128
ipv6 eigrp 100
interface GigabitEthernet0/0/0
ip address 10.0.34.2 255.255.255.252
negotiation auto
ipv6 address 2001:DB8:ACAD:34::2/64
ipv6 enable
ipv6 eigrp 100
interface GigabitEthernet0/0/1
ip address 10.0.45.1 255.255.255.252
negotiation auto
ipv6 address 2001:DB8:ACAD:45::1/64
interface GigabitEthernet0
vrf forwarding Mgmt-intf
no ip address
shutdown
negotiation auto
router eigrp 100
network 4.4.4.4 0.0.0.0
network 10.0.34.0 0.0.0.3
redistribute bgp 65002 metric 100 1 255 1 1500
redistribute connected
router bgp 65002
bgp router-id 4.4.4.4
bgp log-neighbor-changes
neighbor 10.0.45.2 remote-as 65003
neighbor 2001:DB8:ACAD:45::2 remote-as 65003
address-family ipv4
    network 10.0.45.0 mask 255.255.255.252
    redistribute connected
    redistribute eigrp 100 metric 10
```

```
neighbor 10.0.45.2 activate
neighbor 10.0.45.2 next-hop-self
no neighbor 2001:DB8:ACAD:45::2 activate
exit-address-family
address-family ipv6
    redistribute connected
    redistribute eigrp 100 metric 10
    network 2001:DB8:ACAD:45::/64
    neighbor 2001:DB8:ACAD:45::2 activate
    neighbor 2001:DB8:ACAD:45::2 next-hop-self
exit-address-family
ip http server
ip http authentication local
ip http secure-server
ip forward-protocol nd
ip tftp source-interface GigabitEthernet0
ipv6 router eigrp 100
eigrp router-id 4.4.4.4
redistribute connected
redistribute bgp 65002 metric 10000 100 255 1 1500
control-plane
mgcp behavior rsip-range tgcp-only
mgcp behavior comedia-role none
mgcp behavior comedia-check-media-src disable
mgcp behavior comedia-sdp-force disable
mgcp profile default
line con 0
stopbits 1
line aux 0
line vty 0 4
login
transport input ssh
line vty 5 14
login
```

```
transport input ssh
```

```
call-home
```

```
If contact email address in call-home is configured as sch-smart-licensing@cisco.com
```

```
the email address configured in Cisco Smart License Portal will be used as contact email address to send SCH notifications.
```

```
contact-email-addr sch-smart-licensing@cisco.com
```

```
profile "CiscoTAC-1"
```

```
active
```

```
destination transport-method http
```

```
end
```

R5:

```
interface Loopback0
```

```
ip address 5.5.5.5 255.255.255.255
```

```
ip router isis CORE
```

```
ipv6 address 2001:DB8::5/128
```

```
ipv6 router isis CORE
```

```
interface GigabitEthernet0/0/0
```

```
ip address 10.0.56.1 255.255.255.252
```

```
ip router isis CORE
```

```
negotiation auto
```

```
ipv6 address 2001:DB8:ACAD:56::1/64
```

```
ipv6 enable
```

```
ipv6 router isis CORE
```

```
interface GigabitEthernet0/0/1
```

```
ip address 10.0.45.2 255.255.255.252
```

```
negotiation auto
```

```
ipv6 address 2001:DB8:ACAD:45::2/64
```

```
interface GigabitEthernet0/1/0
```

```
negotiation auto
```

```
interface GigabitEthernet0/1/1
```

```
negotiation auto
```

```
interface GigabitEthernet0
```

```
vrf forwarding Mgmt-intf
```

```
no ip address
```

```
shutdown
negotiation auto
interface Vlan1
no ip address
shutdown
router isis CORE
net 49.0001.0000.0000.0003.00
is-type level-2-only
redistribute connected metric 10
redistribute bgp 65003 metric 10
address-family ipv6
    redistribute connected metric 10
    redistribute bgp 65003 metric 10
exit-address-family
router bgp 65003
bgp log-neighbor-changes
bgp default local-preference 200
neighbor 10.0.45.1 remote-as 65002
neighbor 2001:DB8:ACAD:45::1 remote-as 65002
address-family ipv4
    network 10.0.45.0 mask 255.255.255.252
    redistribute connected metric 10
    redistribute isis CORE level-2 metric 10
    neighbor 10.0.45.1 activate
    no neighbor 2001:DB8:ACAD:45::1 activate
exit-address-family

address-family ipv6
    redistribute connected metric 10
    redistribute isis CORE metric 10 level-2
    network 2001:DB8:ACAD:45::/64
    neighbor 2001:DB8:ACAD:45::1 activate
    neighbor 2001:DB8:ACAD:45::1 next-hop-self
exit-address-family
```



```
ip forward-protocol nd
ip http server
ip http authentication local
ip http secure-server
ip tftp source-interface GigabitEthernet0
ip prefix-list net4 seq 5 permit 6.6.6.6/32
route-map PREF permit 10
match ip address prefix-list net4
set local-preference 300
control-plane
line con 0
stopbits 1
line aux 0
stopbits 1
line vty 0 4
login
transport input ssh
line vty 5 14
login
transport input ssh
end
```

R6:

```
interface Loopback0
ip address 6.6.6.6 255.255.255.255
ip router isis CORE
ipv6 address 2001:DB8:ACAD::6/128
ipv6 enable
ipv6 router isis CORE
interface GigabitEthernet0/0/0
ip address 10.0.56.2 255.255.255.252
ip router isis CORE
negotiation auto
ipv6 address 2001:DB8:ACAD:56::2/64
ipv6 enable
```

```
ipv6 router isis CORE
interface GigabitEthernet0/0/1
no ip address
negotiation auto
interface Serial0/1/0
no ip address
shutdown
```

```
interface Serial0/1/1
no ip address
shutdown
```

```
interface GigabitEthernet0
vrf forwarding Mgmt-intf
no ip address
shutdown
negotiation auto
```

```
router isis CORE
net 49.0001.0000.0000.0002.00
is-type level-2-only
metric-style narrow
```

```
router bgp 65003
bgp log-neighbor-changes
bgp default local-preference 200
neighbor 10.0.56.1 remote-as 65003
```

```
address-family ipv4
  neighbor 10.0.56.1 activate
exit-address-family
```

```
ip forward-protocol nd
no ip http server
```

```
ip http authentication local
no ip http secure-server
ip tftp source-interface GigabitEthernet0
```

```
control-plane
```

```
line con 0
transport input none
stopbits 1
line aux 0
stopbits 1
line vty 0 4
login
```

Lab Show Commands:

R4:

```
R4#show ip route
```

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, m - OMP
n - NAT, Ni - NAT inside, No - NAT outside, Nd - NAT DIA
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS

level-2

ia - IS-IS inter area, * - candidate default, U - per-user static

route

H - NHRP, G - NHRP registered, g - NHRP registration summary
o - ODR, P - periodic downloaded static route, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from

PfR

& - replicated local route overrides by connected

Gateway of last resort is not set

1.0.0.0/32 is subnetted, 1 subnets

D EX 1.1.1.1 [170/25600512] via 10.0.34.1, 00:21:53,
GigabitEthernet0/0/0

2.0.0.0/32 is subnetted, 1 subnets

D EX 2.2.2.2 [170/25600512] via 10.0.34.1, 00:21:58,
GigabitEthernet0/0/0

3.0.0.0/32 is subnetted, 1 subnets

D 3.3.3.3 [90/130816] via 10.0.34.1, 00:21:58,
GigabitEthernet0/0/0

4.0.0.0/32 is subnetted, 1 subnets

```

C      4.4.4.4 is directly connected, Loopback0
      5.0.0.0/32 is subnetted, 1 subnets
B      5.5.5.5 [20/10] via 10.0.45.2, 00:21:45
      6.0.0.0/32 is subnetted, 1 subnets
B      6.6.6.6 [20/10] via 10.0.45.2, 00:21:14
      10.0.0.0/8 is variably subnetted, 7 subnets, 2 masks
D EX   10.0.12.0/30
      [170/25600512] via 10.0.34.1, 00:21:58, GigabitEthernet0/0/0
D EX   10.0.23.0/30 [170/3072] via 10.0.34.1, 00:21:58,
GigabitEthernet0/0/0
C      10.0.34.0/30 is directly connected, GigabitEthernet0/0/0
L      10.0.34.2/32 is directly connected, GigabitEthernet0/0/0
C      10.0.45.0/30 is directly connected, GigabitEthernet0/0/1
L      10.0.45.1/32 is directly connected, GigabitEthernet0/0/1
B      10.0.56.0/30 [20/10] via 10.0.45.2, 00:21:45
R4#show ipv6 route
IPv6 Routing Table - default - 14 entries
Codes: C - Connected, L - Local, S - Static, U - Per-user Static route
      B - BGP, R - RIP, H - NHRP, I1 - ISIS L1
      I2 - ISIS L2, IA - ISIS interarea, IS - ISIS summary, D - EIGRP
      EX - EIGRP external, ND - ND Default, NDp - ND Prefix, DCE -
Destination
      NDr - Redirect, O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1
      OE2 - OSPF ext 2, ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2
      a - Application, m - OMP
B      2001:DB8::5/128 [20/10], tag 65003
      via FE80::CE7F:76FF:FE83:1401, GigabitEthernet0/0/1
EX      2001:DB8:ACAD::1/128 [170/281856]
      via FE80::227:90FF:FED5:FAD0, GigabitEthernet0/0/0
EX      2001:DB8:ACAD::2/128 [170/281856]
      via FE80::227:90FF:FED5:FAD0, GigabitEthernet0/0/0
D      2001:DB8:ACAD::3/128 [90/130816]
      via FE80::227:90FF:FED5:FAD0, GigabitEthernet0/0/0
LC      2001:DB8:ACAD::4/128 [0/0]
      via Loopback0, receive
B      2001:DB8:ACAD::6/128 [20/10], tag 65003
      via FE80::CE7F:76FF:FE83:1401, GigabitEthernet0/0/1
EX      2001:DB8:ACAD:12::/64 [170/281856]
      via FE80::227:90FF:FED5:FAD0, GigabitEthernet0/0/0
EX      2001:DB8:ACAD:23::/64 [170/3072]
      via FE80::227:90FF:FED5:FAD0, GigabitEthernet0/0/0
C      2001:DB8:ACAD:34::/64 [0/0]
      via GigabitEthernet0/0/0, directly connected
L      2001:DB8:ACAD:34::2/128 [0/0]
      via GigabitEthernet0/0/0, receive
C      2001:DB8:ACAD:45::/64 [0/0]
      via GigabitEthernet0/0/1, directly connected
L      2001:DB8:ACAD:45::1/128 [0/0]
      via GigabitEthernet0/0/1, receive
B      2001:DB8:ACAD:56::/64 [20/10], tag 65003
      via FE80::CE7F:76FF:FE83:1401, GigabitEthernet0/0/1

```

```

L   FF00::/8 [0/0]
    via Null0, receive
R4#show ip bgp
BGP table version is 12, local router ID is 4.4.4.4
Status codes: s suppressed, d damped, h history, * valid, > best, i -
internal,
             r RIB-failure, S Stale, m multipath, b backup-path, f RT-
Filter,
             x best-external, a additional-path, c RIB-compressed,
             t secondary path, L long-lived-stale,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

```

	Network	Next Hop	Metric	LocPrf	Weight	Path
*>	1.1.1.1/32	10.0.34.1	10		32768	?
*>	2.2.2.2/32	10.0.34.1	10		32768	?
*>	3.3.3.3/32	10.0.34.1	10		32768	?
*>	4.4.4.4/32	0.0.0.0	0		32768	?
*>	5.5.5.5/32	10.0.45.2	10		0	65003 ?
*>	6.6.6.6/32	10.0.45.2	10		0	65003 ?
*>	10.0.12.0/30	10.0.34.1	10		32768	?
*>	10.0.23.0/30	10.0.34.1	10		32768	?
*>	10.0.34.0/30	0.0.0.0	0		32768	?
*	10.0.45.0/30	10.0.45.2	0		0	65003 i
*>		0.0.0.0	0		32768	i
*>	10.0.56.0/30	10.0.45.2	10		0	65003 ?

```

R4#show bgp ipv6
% Command accepted but obsolete, unreleased or unsupported; see
documentation.

```

```

BGP table version is 12, local router ID is 4.4.4.4
Status codes: s suppressed, d damped, h history, * valid, > best, i -
internal,
             r RIB-failure, S Stale, m multipath, b backup-path, f RT-
Filter,
             x best-external, a additional-path, c RIB-compressed,
             t secondary path, L long-lived-stale,
Origin codes: i - IGP, e - EGP, ? - incomplete
RPKI validation codes: V valid, I invalid, N Not found

```

	Network	Next Hop	Metric	LocPrf	Weight	Path
*>	2001:DB8::5/128	2001:DB8:ACAD:45::2				
			10		0	65003 ?
*>	2001:DB8:ACAD::1/128	FE80::227:90FF:FED5:FAD0				
			10		32768	?
*>	2001:DB8:ACAD::2/128	FE80::227:90FF:FED5:FAD0				
			10		32768	?
*>	2001:DB8:ACAD::3/128	FE80::227:90FF:FED5:FAD0				

		10	32768 ?
*>	2001:DB8:ACAD::4/128		
	::	0	32768 ?
*>	2001:DB8:ACAD::6/128		
	2001:DB8:ACAD:45::2		
		10	0 65003 ?
*>	2001:DB8:ACAD:12::/64		
	FE80::227:90FF:FED5:FAD0		
		10	32768 ?
*>	2001:DB8:ACAD:23::/64		
	FE80::227:90FF:FED5:FAD0		
		10	32768 ?
*>	2001:DB8:ACAD:34::/64		
	::	0	32768 ?
*	2001:DB8:ACAD:45::/64		
	2001:DB8:ACAD:45::2		
		10	0 65003 i
*>	::	0	32768 i
*>	2001:DB8:ACAD:56::/64		
	2001:DB8:ACAD:45::2		
		10	0 65003 ?

```

R4#show ip eigrp neighbors
EIGRP-IPv4 Neighbors for AS(100)
H   Address                Interface      Hold Uptime    SRTT
RTO  Q  Seq                                     (sec)         (ms)

Cnt Num
0   10.0.34.1              Gi0/0/0       12 00:24:41   814
4884  0  7

```

```

R4#show ipv6 eigrp neighbors
EIGRP-IPv6 Neighbors for AS(100)
H   Address                Interface      Hold Uptime    SRTT
RTO  Q  Seq                                     (sec)         (ms)

Cnt Num
0   Link-local address:    Gi0/0/0       10 00:24:48   652
3912  0  8

```

```

FE80::227:90FF:FED5:FAD0

```

```

R4#show ip eigrp topology
EIGRP-IPv4 Topology Table for AS(100)/ID(4.4.4.4)
Codes: P - Passive, A - Active, U - Update, Q - Query, R - Reply,
       r - reply Status, s - sia Status

```

```

P 10.0.34.0/30, 1 successors, FD is 2816
    via Connected, GigabitEthernet0/0/0
P 4.4.4.4/32, 1 successors, FD is 128256
    via Connected, Loopback0
P 10.0.56.0/30, 1 successors, FD is 25600256, tag is 65003
    via Redistributed (25600256/0)
P 5.5.5.5/32, 1 successors, FD is 25600256, tag is 65003
    via Redistributed (25600256/0)

```

P 2.2.2.2/32, 1 successors, FD is 25600512, tag is 65001
via 10.0.34.1 (25600512/25600256), GigabitEthernet0/0/0
P 6.6.6.6/32, 1 successors, FD is 25600256, tag is 65003
via Redistributed (25600256/0)
P 10.0.45.0/30, 1 successors, FD is 2816
via Rconnected (2816/0)
P 10.0.23.0/30, 1 successors, FD is 3072
via 10.0.34.1 (3072/2816), GigabitEthernet0/0/0
P 3.3.3.3/32, 1 successors, FD is 130816
via 10.0.34.1 (130816/128256), GigabitEthernet0/0/0
P 1.1.1.1/32, 1 successors, FD is 25600512, tag is 65001
via 10.0.34.1 (25600512/25600256), GigabitEthernet0/0/0
P 10.0.12.0/30, 1 successors, FD is 25600512, tag is 65001
via 10.0.34.1 (25600512/25600256), GigabitEthernet0/0/0

R4#show ipv6 eigrp topology

EIGRP-IPv6 Topology Table for AS(100)/ID(4.4.4.4)

Codes: P - Passive, A - Active, U - Update, Q - Query, R - Reply,
r - reply Status, s - sia Status

P 2001:DB8::5/128, 1 successors, FD is 281600, tag is 65003
via Redistributed (281600/0)
P 2001:DB8:ACAD:45::/64, 1 successors, FD is 2816
via Rconnected (2816/0)
P 2001:DB8:ACAD:56::/64, 1 successors, FD is 281600, tag is 65003
via Redistributed (281600/0)
P 2001:DB8:ACAD::2/128, 1 successors, FD is 281856
via FE80::227:90FF:FED5:FAD0 (281856/281600),
GigabitEthernet0/0/0
P 2001:DB8:ACAD:12::/64, 1 successors, FD is 281856
via FE80::227:90FF:FED5:FAD0 (281856/281600),
GigabitEthernet0/0/0
P 2001:DB8:ACAD:34::/64, 1 successors, FD is 2816
via Connected, GigabitEthernet0/0/0
P 2001:DB8:ACAD::1/128, 1 successors, FD is 281856
via FE80::227:90FF:FED5:FAD0 (281856/281600),
GigabitEthernet0/0/0
P 2001:DB8:ACAD::3/128, 1 successors, FD is 130816
via FE80::227:90FF:FED5:FAD0 (130816/128256),
GigabitEthernet0/0/0
P 2001:DB8:ACAD::4/128, 1 successors, FD is 128256
via Connected, Loopback0
P 2001:DB8:ACAD::6/128, 1 successors, FD is 281600, tag is 65003
via Redistributed (281600/0)
P 2001:DB8:ACAD:23::/64, 1 successors, FD is 3072
via FE80::227:90FF:FED5:FAD0 (3072/2816), GigabitEthernet0/0/0

R5:

R5#show ip route

*Nov 17 17:55:54.649: %SYS-5-CONFIG_I: Configured from console by console

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area

N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2

E1 - OSPF external type 1, E2 - OSPF external type 2

i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS

level-2

ia - IS-IS inter area, * - candidate default, U - per-user static

route

o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP

a - application route

+ - replicated route, % - next hop override, p - overrides from

PfR

Gateway of last resort is not set

1.0.0.0/32 is subnetted, 1 subnets

B 1.1.1.1 [20/10] via 10.0.45.1, 00:16:43

2.0.0.0/32 is subnetted, 1 subnets

B 2.2.2.2 [20/10] via 10.0.45.1, 00:16:43

3.0.0.0/32 is subnetted, 1 subnets

B 3.3.3.3 [20/10] via 10.0.45.1, 00:16:43

4.0.0.0/32 is subnetted, 1 subnets

B 4.4.4.4 [20/0] via 10.0.45.1, 00:16:43

5.0.0.0/32 is subnetted, 1 subnets

C 5.5.5.5 is directly connected, Loopback0

6.0.0.0/32 is subnetted, 1 subnets

i L2 6.6.6.6 [115/20] via 10.0.56.2, 00:16:32, GigabitEthernet0/0/0

10.0.0.0/8 is variably subnetted, 7 subnets, 2 masks

B 10.0.12.0/30 [20/10] via 10.0.45.1, 00:16:43

B 10.0.23.0/30 [20/10] via 10.0.45.1, 00:16:43

B 10.0.34.0/30 [20/0] via 10.0.45.1, 00:16:43

C 10.0.45.0/30 is directly connected, GigabitEthernet0/0/1

L 10.0.45.2/32 is directly connected, GigabitEthernet0/0/1

C 10.0.56.0/30 is directly connected, GigabitEthernet0/0/0

L 10.0.56.1/32 is directly connected, GigabitEthernet0/0/0

R5#show ipv6 route

IPv6 Routing Table - default - 14 entries

Codes: C - Connected, L - Local, S - Static, U - Per-user Static route

B - BGP, R - RIP, I1 - ISIS L1, I2 - ISIS L2

IA - ISIS interarea, IS - ISIS summary, D - EIGRP, EX - EIGRP
external

ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr -
Redirect

O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1, OE2 - OSPF ext
2

ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2, a - Application

LC 2001:DB8::5/128 [0/0]

via Loopback0, receive


```

B    2001:DB8:ACAD::1/128 [20/10]
    via FE80::DEF7:19FF:FE78:4891, GigabitEthernet0/0/1
B    2001:DB8:ACAD::2/128 [20/10]
    via FE80::DEF7:19FF:FE78:4891, GigabitEthernet0/0/1
B    2001:DB8:ACAD::3/128 [20/10]
    via FE80::DEF7:19FF:FE78:4891, GigabitEthernet0/0/1
B    2001:DB8:ACAD::4/128 [20/0]
    via FE80::DEF7:19FF:FE78:4891, GigabitEthernet0/0/1
I2   2001:DB8:ACAD::6/128 [115/20]
    via FE80::521C:B0FF:FE42:E970, GigabitEthernet0/0/0
B    2001:DB8:ACAD:12::/64 [20/10]
    via FE80::DEF7:19FF:FE78:4891, GigabitEthernet0/0/1
B    2001:DB8:ACAD:23::/64 [20/10]
    via FE80::DEF7:19FF:FE78:4891, GigabitEthernet0/0/1
B    2001:DB8:ACAD:34::/64 [20/0]
    via FE80::DEF7:19FF:FE78:4891, GigabitEthernet0/0/1
C    2001:DB8:ACAD:45::/64 [0/0]
    via GigabitEthernet0/0/1, directly connected
L    2001:DB8:ACAD:45::2/128 [0/0]
    via GigabitEthernet0/0/1, receive
C    2001:DB8:ACAD:56::/64 [0/0]
    via GigabitEthernet0/0/0, directly connected
L    2001:DB8:ACAD:56::1/128 [0/0]
    via GigabitEthernet0/0/0, receive
L    FF00::/8 [0/0]
    via Null0, receive
R5#show isis * rib

```

IPv4 local RIB for IS-IS process CORE

IPv4 unicast topology base (TID 0, TOPOID 0x0) =====

6.6.6.6/32

```

[115/L2/20] via 10.0.56.2(GigabitEthernet0/0/0), from 6.6.6.6, tag 0,
LSP[2/2]
    (installed)

```

10.0.56.0/30

```

[115/L2/20] via 10.0.56.2(GigabitEthernet0/0/0), from 6.6.6.6, tag 0,
LSP[2/2]

```

IS-IS IPv6 process CORE, local RIB

```

* 2001:DB8:ACAD::6/128
    via FE80::521C:B0FF:FE42:E970/GigabitEthernet0/0/0, type L2 metric
20 tag 0 LSP [2/2]
    2001:DB8:ACAD:56::/64
    via FE80::521C:B0FF:FE42:E970/GigabitEthernet0/0/0, type L2 metric
20 tag 0 LSP [2/2]

```

R5# show isis * topology

Tag CORE:

IS-IS TID 0 paths to level-2 routers

System Id	Metric	Next-Hop	Interface	SNPA
R6	10	R6	Gi0/0/0	
501c.b042.e970				
R5	--			

R5#show isis neighbors

Tag CORE:

System Id	Type	Interface	IP Address	State	Holdtime	Circuit
R6	L2	Gi0/0/0	10.0.56.2	UP	28	R5.01

R5#show ip bgp

BGP table version is 12, local router ID is 5.5.5.5

Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,

r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,

x best-external, a additional-path, c RIB-compressed,

Origin codes: i - IGP, e - EGP, ? - incomplete

RPKI validation codes: V valid, I invalid, N Not found

	Network	Next Hop	Metric	LocPrf	Weight	Path
*>	1.1.1.1/32	10.0.45.1	10		0	65002 ?
*>	2.2.2.2/32	10.0.45.1	10		0	65002 ?
*>	3.3.3.3/32	10.0.45.1	10		0	65002 ?
*>	4.4.4.4/32	10.0.45.1	0		0	65002 ?
*>	5.5.5.5/32	0.0.0.0	10		32768	?
*>	6.6.6.6/32	10.0.56.2	10		32768	?
*>	10.0.12.0/30	10.0.45.1	10		0	65002 ?
*>	10.0.23.0/30	10.0.45.1	10		0	65002 ?
*>	10.0.34.0/30	10.0.45.1	0		0	65002 ?
*	10.0.45.0/30	10.0.45.1	0		0	65002

iR4#show ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP

D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area

N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2

E1 - OSPF external type 1, E2 - OSPF external type 2, m - OMP

n - NAT, Ni - NAT inside, No - NAT outside, Nd - NAT DIA

i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS

level-2

ia - IS-IS inter area, * - candidate default, U - per-user static

route

H - NHRP, G - NHRP registered, g - NHRP registration summary

o - ODR, P - periodic downloaded static route, l - LISP

a - application route

+ - replicated route, % - next hop override, p - overrides from

PfR

& - replicated local route overrides by connected

Gateway of last resort is not set

```
1.0.0.0/32 is subnetted, 1 subnets
D EX    1.1.1.1 [170/25600512] via 10.0.34.1, 00:21:53,
GigabitEthernet0/0/0
2.0.0.0/32 is subnetted, 1 subnets
D EX    2.2.2.2 [170/25600512] via 10.0.34.1, 00:21:58,
GigabitEthernet0/0/0
3.0.0.0/32 is subnetted, 1 subnets
D       3.3.3.3 [90/130816] via 10.0.34.1, 00:21:58,
GigabitEthernet0/0/0
4.0.0.0/32 is subnetted, 1 subnets
C       4.4.4.4 is directly connected, Loopback0
5.0.0.0/32 is subnetted, 1 subnets
B       5.5.5.5 [20/10] via 10.0.45.2, 00:21:45
6.0.0.0/32 is subnetted, 1 subnets
B       6.6.6.6 [20/10] via 10.0.45.2, 00:21:14
10.0.0.0/8 is variably subnetted, 7 subnets, 2 masks
D EX    10.0.12.0/30
        [170/25600512] via 10.0.34.1, 00:21:58, GigabitEthernet0/0/0
D EX    10.0.23.0/30 [170/3072] via 10.0.34.1, 00:21:58,
GigabitEthernet0/0/0
C       10.0.34.0/30 is directly connected, GigabitEthernet0/0/0
L       10.0.34.2/32 is directly connected, GigabitEthernet0/0/0
C       10.0.45.0/30 is directly connected, GigabitEthernet0/0/1
L       10.0.45.1/32 is directly connected, GigabitEthernet0/0/1
B       10.0.56.0/30 [20/10] via 10.0.45.2, 00:21:45
```

R4#show ipv6 route

IPv6 Routing Table - default - 14 entries

Codes: C - Connected, L - Local, S - Static, U - Per-user Static route

B - BGP, R - RIP, H - NHRP, I1 - ISIS L1

I2 - ISIS L2, IA - ISIS interarea, IS - ISIS summary, D - EIGRP

EX - EIGRP external, ND - ND Default, NDp - ND Prefix, DCE -

Destination

NDr - Redirect, O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1

OE2 - OSPF ext 2, ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2

a - Application, m - OMP

```
B 2001:DB8::5/128 [20/10], tag 65003
  via FE80::CE7F:76FF:FE83:1401, GigabitEthernet0/0/1
EX 2001:DB8:ACAD::1/128 [170/281856]
  via FE80::227:90FF:FED5:FAD0, GigabitEthernet0/0/0
EX 2001:DB8:ACAD::2/128 [170/281856]
  via FE80::227:90FF:FED5:FAD0, GigabitEthernet0/0/0
D 2001:DB8:ACAD::3/128 [90/130816]
  via FE80::227:90FF:FED5:FAD0, GigabitEthernet0/0/0
LC 2001:DB8:ACAD::4/128 [0/0]
  via Loopback0, receive
B 2001:DB8:ACAD::6/128 [20/10], tag 65003
  via FE80::CE7F:76FF:FE83:1401, GigabitEthernet0/0/1
EX 2001:DB8:ACAD:12::/64 [170/281856]
  via FE80::227:90FF:FED5:FAD0, GigabitEthernet0/0/0
```

```

EX 2001:DB8:ACAD:23::/64 [170/3072]
    via FE80::227:90FF:FED5:FAD0, GigabitEthernet0/0/0
C 2001:DB8:ACAD:34::/64 [0/0]
    via GigabitEthernet0/0/0, directly connected
L 2001:DB8:ACAD:34::2/128 [0/0]
    via GigabitEthernet0/0/0, receive
C 2001:DB8:ACAD:45::/64 [0/0]
    via GigabitEthernet0/0/1, directly connected
L 2001:DB8:ACAD:45::1/128 [0/0]
    via GigabitEthernet0/0/1, receive
B 2001:DB8:ACAD:56::/64 [20/10], tag 65003
    via FE80::CE7F:76FF:FE83:1401, GigabitEthernet0/0/1
L FF00::/8 [0/0]
    via Null0, receive

```

```

*> 0.0.0.0 0 32768 i
*> 10.0.56.0/30 0.0.0.0 10 32768 ?

```

R5#show bgp ipv6 unicast

BGP table version is 20, local router ID is 5.5.5.5

Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,

r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter,

x best-external, a additional-path, c RIB-compressed,

Origin codes: i - IGP, e - EGP, ? - incomplete

RPKI validation codes: V valid, I invalid, N Not found

	Network	Next Hop	Metric	LocPrf	Weight	Path
*>	2001:DB8::5/128	::	10		32768	?
*>	2001:DB8:ACAD::1/128	2001:DB8:ACAD:45::1	10		0	65002 ?
*>	2001:DB8:ACAD::2/128	2001:DB8:ACAD:45::1	10		0	65002 ?
*>	2001:DB8:ACAD::3/128	2001:DB8:ACAD:45::1	10		0	65002 ?
*>	2001:DB8:ACAD::4/128	2001:DB8:ACAD:45::1	0		0	65002 ?
*>	2001:DB8:ACAD::6/128	FE80::521C:B0FF:FE42:E970	10		32768	?
*>	2001:DB8:ACAD:12::/64	2001:DB8:ACAD:45::1	10		0	65002 ?
*>	2001:DB8:ACAD:23::/64	2001:DB8:ACAD:45::1	10		0	65002 ?
*>	2001:DB8:ACAD:34::/64					

	2001:DB8:ACAD:45::1	0	0 65002 ?
*	2001:DB8:ACAD:45::/64		
	2001:DB8:ACAD:45::1	0	0 65002 i
*>	::	10	32768 i
*>	2001:DB8:ACAD:56::/64		
	::	10	32768 ?

```

R6:
R6(config)#do traceroute 1.1.1.1
Type escape sequence to abort.
Tracing the route to 1.1.1.1
VRF info: (vrf in name/id, vrf out name/id)
  1 10.0.56.1 1 msec 1 msec 0 msec
  2 10.0.45.1 1 msec 1 msec 0 msec
  3 10.0.34.1 1 msec 2 msec 1 msec
  4 10.0.23.1 1 msec 2 msec 1 msec
  5 10.0.12.1 2 msec 2 msec *
R6(config)#do traceroute 2001:db8:acad::1
Type escape sequence to abort.
Tracing the route to 2001:DB8:ACAD::1

  1 2001:DB8:ACAD:56::1 6 msec 1 msec 0 msec
  2 2001:DB8:ACAD:45::1 14 msec 1 msec 0 msec
  3 2001:DB8:ACAD:34::1 1 msec 2 msec 1 msec
  4 2001:DB8:ACAD:23::1 1 msec 9 msec 2 msec
  5 2001:DB8:ACAD:12::1 1 msec 2 msec 2 msec
R6#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSR4#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, m - OMP
       n - NAT, Ni - NAT inside, No - NAT outside, Nd - NAT DIA
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS
level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static
route
       H - NHRP, G - NHRP registered, g - NHRP registration summary
       o - ODR, P - periodic downloaded static route, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from
PFR
       & - replicated local route overrides by connected

Gateway of last resort is not set

    1.0.0.0/32 is subnetted, 1 subnets
D EX    1.1.1.1 [170/25600512] via 10.0.34.1, 00:21:53,
GigabitEthernet0/0/0
    2.0.0.0/32 is subnetted, 1 subnets
D EX    2.2.2.2 [170/25600512] via 10.0.34.1, 00:21:58,
GigabitEthernet0/0/0
    3.0.0.0/32 is subnetted, 1 subnets
D       3.3.3.3 [90/130816] via 10.0.34.1, 00:21:58,
GigabitEthernet0/0/0
    4.0.0.0/32 is subnetted, 1 subnets

```

```

C      4.4.4.4 is directly connected, Loopback0
      5.0.0.0/32 is subnetted, 1 subnets
B      5.5.5.5 [20/10] via 10.0.45.2, 00:21:45
      6.0.0.0/32 is subnetted, 1 subnets
B      6.6.6.6 [20/10] via 10.0.45.2, 00:21:14
      10.0.0.0/8 is variably subnetted, 7 subnets, 2 masks
D EX   10.0.12.0/30
      [170/25600512] via 10.0.34.1, 00:21:58, GigabitEthernet0/0/0
D EX   10.0.23.0/30 [170/3072] via 10.0.34.1, 00:21:58,
GigabitEthernet0/0/0
C      10.0.34.0/30 is directly connected, GigabitEthernet0/0/0
L      10.0.34.2/32 is directly connected, GigabitEthernet0/0/0
C      10.0.45.0/30 is directly connected, GigabitEthernet0/0/1
L      10.0.45.1/32 is directly connected, GigabitEthernet0/0/1
B      10.0.56.0/30 [20/10] via 10.0.45.2, 00:21:45
R4#show ipv6 route
IPv6 Routing Table - default - 14 entries
Codes: C - Connected, L - Local, S - Static, U - Per-user Static route
      B - BGP, R - RIP, H - NHRP, I1 - ISIS L1
      I2 - ISIS L2, IA - ISIS interarea, IS - ISIS summary, D - EIGRP
      EX - EIGRP external, ND - ND Default, NDp - ND Prefix, DCE -
Destination
      NDr - Redirect, O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1
      OE2 - OSPF ext 2, ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2
      a - Application, m - OMP
B      2001:DB8::5/128 [20/10], tag 65003
      via FE80::CE7F:76FF:FE83:1401, GigabitEthernet0/0/1
EX      2001:DB8:ACAD::1/128 [170/281856]
      via FE80::227:90FF:FED5:FAD0, GigabitEthernet0/0/0
EX      2001:DB8:ACAD::2/128 [170/281856]
      via FE80::227:90FF:FED5:FAD0, GigabitEthernet0/0/0
D      2001:DB8:ACAD::3/128 [90/130816]
      via FE80::227:90FF:FED5:FAD0, GigabitEthernet0/0/0
LC      2001:DB8:ACAD::4/128 [0/0]
      via Loopback0, receive
B      2001:DB8:ACAD::6/128 [20/10], tag 65003
      via FE80::CE7F:76FF:FE83:1401, GigabitEthernet0/0/1
EX      2001:DB8:ACAD:12::/64 [170/281856]
      via FE80::227:90FF:FED5:FAD0, GigabitEthernet0/0/0
EX      2001:DB8:ACAD:23::/64 [170/3072]
      via FE80::227:90FF:FED5:FAD0, GigabitEthernet0/0/0
C      2001:DB8:ACAD:34::/64 [0/0]
      via GigabitEthernet0/0/0, directly connected
L      2001:DB8:ACAD:34::2/128 [0/0]
      via GigabitEthernet0/0/0, receive
C      2001:DB8:ACAD:45::/64 [0/0]
      via GigabitEthernet0/0/1, directly connected
L      2001:DB8:ACAD:45::1/128 [0/0]
      via GigabitEthernet0/0/1, receive
B      2001:DB8:ACAD:56::/64 [20/10], tag 65003
      via FE80::CE7F:76FF:FE83:1401, GigabitEthernet0/0/1

```

```

L   FF00::/8 [0/0]
    via Null0, receiveA external type 1, N2 - OSPF NSSA external type 2
    E1 - OSPF external type 1, E2 - OSPF external type 2
    i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS
level-2
    ia - IS-IS inter area, * - candidate default, U - per-user static
route
    o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
    a - application route
    + - replicated route, % - next hop override, p - overrides from
PfR

```

Gateway of last resort is not set

```

    1.0.0.0/32 is subnetted, 1 subnets
i L2   1.1.1.1 [115/20] via 10.0.56.1, 00:07:00, GigabitEthernet0/0/0
    2.0.0.0/32 is subnetted, 1 subnets
i L2   2.2.2.2 [115/20] via 10.0.56.1, 00:07:00, GigabitEthernet0/0/0
    3.0.0.0/32 is subnetted, 1 subnets
i L2   3.3.3.3 [115/20] via 10.0.56.1, 00:07:00, GigabitEthernet0/0/0
    4.0.0.0/32 is subnetted, 1 subnets
i L2   4.4.4.4 [115/20] via 10.0.56.1, 00:07:00, GigabitEthernet0/0/0
    5.0.0.0/32 is subnetted, 1 subnets
i L2   5.5.5.5 [115/20] via 10.0.56.1, 00:07:00, GigabitEthernet0/0/0
    6.0.0.0/32 is subnetted, 1 subnets
C       6.6.6.6 is directly connected, Loopback0
    10.0.0.0/8 is variably subnetted, 6 subnets, 2 masks
i L2   10.0.12.0/30 [115/20] via 10.0.56.1, 00:07:00,
GigabitEthernet0/0/0
i L2   10.0.23.0/30 [115/20] via 10.0.56.1, 00:07:00,
GigabitEthernet0/0/0
i L2   10.0.34.0/30 [115/20] via 10.0.56.1, 00:07:00,
GigabitEthernet0/0/0
i L2   10.0.45.0/30 [115/20] via 10.0.56.1, 00:07:00,
GigabitEthernet0/0/0
C       10.0.56.0/30 is directly connected, GigabitEthernet0/0/0
L       10.0.56.2/32 is directly connected, GigabitEthernet0/0/0

```

R6#show ipv6 route

IPv6 Routing Table - default - 13 entries

Codes: C - Connected, L - Local, S - Static, U - Per-user Static route

B - BGP, R - RIP, I1 - ISIS L1, I2 - ISIS L2

IA - ISIS interarea, IS - ISIS summary, D - EIGRP, EX - EIGRP

external

ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr -

Redirect

O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1, OE2 - OSPF ext

2

ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2, a - Application

I2 2001:DB8::5/128 [115/20]


```

    via FE80::CE7F:76FF:FE83:1400, GigabitEthernet0/0/0
I2  2001:DB8:ACAD::1/128 [115/20]
    via FE80::CE7F:76FF:FE83:1400, GigabitEthernet0/0/0
I2  2001:DB8:ACAD::2/128 [115/20]
    via FE80::CE7F:76FF:FE83:1400, GigabitEthernet0/0/0
I2  2001:DB8:ACAD::3/128 [115/20]
    via FE80::CE7F:76FF:FE83:1400, GigabitEthernet0/0/0
I2  2001:DB8:ACAD::4/128 [115/20]
    via FE80::CE7F:76FF:FE83:1400, GigabitEthernet0/0/0
LC  2001:DB8:ACAD::6/128 [0/0]
    via Loopback0, receive
I2  2001:DB8:ACAD:12::/64 [115/20]
    via FE80::CE7F:76FF:FE83:1400, GigabitEthernet0/0/0
I2  2001:DB8:ACAD:23::/64 [115/20]
    via FE80::CE7F:76FF:FE83:1400, GigabitEthernet0/0/0
I2  2001:DB8:ACAD:34::/64 [115/20]
    via FE80::CE7F:76FF:FE83:1400, GigabitEthernet0/0/0
I2  2001:DB8:ACAD:45::/64 [115/20]
    via FE80::CE7F:76FF:FE83:1400, GigabitEthernet0/0/0
C   2001:DB8:ACAD:56::/64 [0/0]
    via GigabitEthernet0/0/0, directly connected
L   2001:DB8:ACAD:56::2/128 [0/0]
    via GigabitEthernet0/0/0, receive
L   FF00::/8 [0/0]
    via Null0, receive
R6#show isis * rib

```

IPv4 local RIB for IS-IS process CORE

IPV4 unicast topology base (TID 0, TOPOID 0x0) =====

```

1.1.1.1/32
  [115/L2/20] via 10.0.56.1(GigabitEthernet0/0/0), from 5.5.5.5, tag 0,
LSP[2/2/4]
              (installed)

2.2.2.2/32
  [115/L2/20] via 10.0.56.1(GigabitEthernet0/0/0), from 5.5.5.5, tag 0,
LSP[2/2/4]
              (installed)

3.3.3.3/32
  [115/L2/20] via 10.0.56.1(GigabitEthernet0/0/0), from 5.5.5.5, tag 0,
LSP[2/2/4]
              (installed)

4.4.4.4/32
  [115/L2/20] via 10.0.56.1(GigabitEthernet0/0/0), from 5.5.5.5, tag 0,
LSP[2/2/4]
              (installed)

```

```

5.5.5.5/32
  [115/L2/20] via 10.0.56.1(GigabitEthernet0/0/0), from 5.5.5.5, tag 0,
  LSP[2/2/4]
      (installed)

10.0.12.0/30
  [115/L2/20] via 10.0.56.1(GigabitEthernet0/0/0), from 5.5.5.5, tag 0,
  LSP[2/2/4]
      (installed)

10.0.23.0/30
  [115/L2/20] via 10.0.56.1(GigabitEthernet0/0/0), from 5.5.5.5, tag 0,
  LSP[2/2/4]
      (installed)

10.0.34.0/30
  [115/L2/20] via 10.0.56.1(GigabitEthernet0/0/0), from 5.5.5.5, tag 0,
  LSP[2/2/4]
      (installed)

10.0.45.0/30
  [115/L2/20] via 10.0.56.1(GigabitEthernet0/0/0), from 5.5.5.5, tag 0,
  LSP[2/2/4]
      (installed)

10.0.56.0/30
  [115/L2/20] via 10.0.56.1(GigabitEthernet0/0/0), from 5.5.5.5, tag 0,
  LSP[2/2/4]
IS-IS IPv6 process CORE, local RIB
* 2001:DB8::5/128
    via FE80::CE7F:76FF:FE83:1400/GigabitEthernet0/0/0, type L2 metric
20 tag 0 LSP [2/4]
* 2001:DB8:ACAD::1/128
    via FE80::CE7F:76FF:FE83:1400/GigabitEthernet0/0/0, type L2 metric
20 tag 0 LSP [2/4]
* 2001:DB8:ACAD::2/128
    via FE80::CE7F:76FF:FE83:1400/GigabitEthernet0/0/0, type L2 metric
20 tag 0 LSP [2/4]
* 2001:DB8:ACAD::3/128
    via FE80::CE7F:76FF:FE83:1400/GigabitEthernet0/0/0, type L2 metric
20 tag 0 LSP [2/4]
* 2001:DB8:ACAD::4/128
    via FE80::CE7F:76FF:FE83:1400/GigabitEthernet0/0/0, type L2 metric
20 tag 0 LSP [2/4]
* 2001:DB8:ACAD:12::/64
    via FE80::CE7F:76FF:FE83:1400/GigabitEthernet0/0/0, type L2 metric
20 tag 0 LSP [2/4]
* 2001:DB8:ACAD:23::/64
    via FE80::CE7F:76FF:FE83:1400/GigabitEthernet0/0/0, type L2 metric
20 tag 0 LSP [2/4]

```

```
* 2001:DB8:ACAD:34::/64
  via FE80::CE7F:76FF:FE83:1400/GigabitEthernet0/0/0, type L2 metric
20 tag 0 LSP [2/4]
* 2001:DB8:ACAD:45::/64
  via FE80::CE7F:76FF:FE83:1400/GigabitEthernet0/0/0, type L2 metric
20 tag 0 LSP [2/4]
  2001:DB8:ACAD:56::/64
    via FE80::CE7F:76FF:FE83:1400/GigabitEthernet0/0/0, type L2 metric
20 tag 0 LSP [2/4]
R6#show isis * topology
```

Tag CORE:

IS-IS TID 0 paths to level-2 routers

System Id	Metric	Next-Hop	Interface	SNPA
R6	--			
R5	10	R5	Gi0/0/0	
cc7f.7683.1400				

R6#show isis neighbors

Tag CORE:

System Id	Type	Interface	IP Address	State	Holdtime	Circuit
R5	L2	Gi0/0/0	10.0.56.1	UP	8	R5.01

Problems:

1. Lack of knowledge in redistribution

Going into this lab, we knew we needed to redistribute routes between our routing protocols. However, we ran into several problems when trying to implement. First of all, we were not aware that each of the edge routers needed to redistribute the routes from their interior gateway protocol into BGP, and BGP needed to be redistributed into the interior gateway protocols. In addition, we spent several weeks unaware that the redistribute commands required manually assigned metric values for the routes to be sent.

2. Lack of knowledge in IS-IS

Before this lab, my group had no experience in IS-IS. This was not as significant a problem as redistribution, as we were able to figure out how to set up IS-IS fairly quickly.

3. Lack of knowledge in EIGRP

Before this lab, my group had very limited experience with EIGRP. While we were able get EIGRP connectivity within an AS, our lack of knowledge about how to redistribute into it led to one of the bigger roadblocks that we encountered, taking us several weeks to resolve.

4. BGP Features

We had to implement two advanced BGP features for our lab. Considering that we had never used BGP before, we had to research which features we could implement, how to implement them, and how to show what impacts they had on our topology. We ended up using weight and AS_PATH, both options that change which router is preferred when routing through BGP.

5. Improper Topology

Before this lab, we did not know how to set up our autonomous systems, so we ended up setting up four when we only needed three. It took us a few weeks before we discovered the proper autonomous system topology and were able to set up our bgp properly.

6. Non-Functional Routers

Several of the routers in our rack were non-functional. Because of this, we had to work around them and wire the lab accordingly.

7. Shared Routers

We shared the routers used in our lab with another group later in the day. Because of this, we had to spend 10-15 minutes rebooting and loading our configs onto the routers every day, eventually wasting hours of time from our working time.

8. Restarting the lab

When doing the lab, we accumulated a large amount of misconfigurations when trying to set up eigrp, is-is, and bgp for the first time. After spending two weeks learning about how to configure each part of the lab, the misconfigurations had accumulated to a point where it no longer made sense to try and debug our lab. Because of this, we decided to restart our lab and redo all of our configurations.